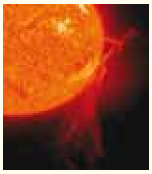
SHORTS

ANTI-GRAVITY TREADMILL

How do you exercise when you're in space and gravity is not weighing you down? A device originally intended to rehabilitate soldiers who have sustained injuries may be the solution. Scientific American reports

http://bit.ly/cdqUGZ

COSMIC GEM, SUN BURP, VEGAS



No they're not names of psychoactive drugs. They're three awesome photographs of anomalies in outer space captured by NASA's space imaging equipment.

http://bit.ly/eECSfs

WIKILEAKS AT IT AGAIN!

Before his arrest Wikileaks spokesperson Julian Assange hinted that the next target would be a major American bank, without naming it. Bank of America has started registering domain names for its senior executives, so that once the documents are leaked the chaos can be contained.

http://rww.to/hOK47C

2011'S MOST ANTICIPATED GEAR

With a host of products lined up for release in 2011, check out what is going to be hot and what could turn out to be the next tech dud. Anticipations are high over the iPad 2 and Sony's PSP phone. We look forward to having a year full of tech innovations and surprises.

http://bit.ly/gx6EfL

The Problem With Big Hard Drives

By Matthew Murray

Ever since hard drives went mainstream in the late 1980s, computer owners have been able to count on one thing: their increasing capacity. As a result, what once used to be little more than a convenient method for reducing disk swapping has now become an integral part of daily life, with hard drives today storing not just our programs and our documents, but our very memories in the form of the photos and videos. The ever-increasing numbers of those have forced storage capacity to skyrocket to still more dizzying levels, to the point where most ordinary people now only really think of it when they're in danger of filling it up. But there's a reason that really large hard drives should be at the forefront of every computer owner's mind: they're about to become unusable.

Okay, perhaps that's a bit hyperbolic—but only a bit. The fact is, we're not just approaching the threshold at which hard drives can't get bigger: We're already there. The advent of very large hard drives, we're talking over 2.19 terabytes (TB) in capacity, in 2010 has highlighted a problem that's been growing for well over a decade, and whose seeds were planted when the very first hard drives hit the market some 25 or 30 years ago. The good news is that there are some solutions to it; the bad news is, they're little more than workarounds for an issue that needs more than makeshift bandages.

What Is the 2.19TB Barrier?

Back in the days when hard drives' capacities were measured in megabytes at the outside (we're talking the 1950s here, though it was still the case well into the 1980s, when floppy disks' sizes were measured only in kilobytes), a block size of 512 bytes made sense as a way of maximizing available space. But during the boom years of the 1990s, when drives were rapidly getting bigger and performance started straining, the 512-byte block size became more restrictive. So the industry decided to up the standard block size to 4,096 bytes, or 4KB (sometimes referred to as "Advanced Format"); 4KB blocks have more efficient methods of error checking, which mean they can devote less space to it, resulting in drives of higher capacities and those that encounter fewer errors.

It was a good idea, but it didn't gain mainstream adoption until after Windows XP made its big splash in 2001. That OS creates primary disk partitions at logical block address 63, just short of being perfectly divisible by eight, so it writes data across both sides of the physical boundary of the 4KB sector. In other words, the world's most popular operating system for the better part of a decade didn't support 4KB blocks, which put a huge crimp in the industry's efforts to expand the initiative. Luckily, there was a